AI for Manufacturing & Supply Chain Institute (AIMS) - Internal Charter Overview

Tecnológico de Monterrey
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Mission

AIMS aims to transform manufacturing and supply chain systems through artificial intelligence, advanced analytics, automation, and materials innovation, generating measurable impact for industry and society.

Vision

By 2030, AIMS is a world-class, industry-connected research institute within Tecnológico de Monterrey, financially diversified beyond tuition subsidies, powered by a fully operational Outreach Membership Model, and globally recognized for advancing Artificial Intelligence for Manufacturing & Supply Chains.

1. Strategic Intent: Targeted Industrial Impact

AIMS focuses on high-impact transformation opportunities at the intersection of AI, manufacturing, materials, and supply chains, where:

- Industrial systems are becoming increasingly data-intensive and autonomous
- Supply chains face systemic disruptions, decarbonization pressure, and complexity
- Manufacturing is undergoing digital and materials-driven transformation

These trends represent multi-trillion-dollar global transitions, with:

- AI in manufacturing projected to exceed $20–30B+ annually in value creation
- Supply chain disruptions costing companies 5–10% of annual revenues
- Decarbonization and Scope 3 pressures reshaping procurement and operations globally

In this context, AIMS focuses on translating these macro-level transformations into targeted, high-impact opportunities for research and deployment. Rather than approaching these domains as broad or independent fields, the AIMS emphasizes their integration, enabling the development of deployable solutions that address real industrial challenges. This positioning reflects both the scale of the opportunity and the need for coordinated multidisciplinary approaches that can generate measurable impact.

2. Research Focus Areas

The research structure of AIMS is organized around four integrated areas: *Smart Manufacturing, Automation & Materials for Manufacturing*, *Artificial Intelligence for Industrial Systems*, *Supply Chain & Logistics Systems*, and *LIFT Lab - AI for Inclusive Supply Chains*.

The area of *Smart Manufacturing, Automation & Materials for Manufacturing* focuses on the convergence of advanced manufacturing processes, robotics, and materials innovation. Its emphasis lies in connecting materials design, process optimization, and system-level performance, enabling the development of next-generation production capabilities. This integration is particularly relevant in the context of sustainable manufacturing, advanced materials, and the increasing role of automation in improving productivity and competitiveness.

The area of *Artificial Intelligence for Industrial Systems* is anchored in advanced technological capabilities and reflects the understanding that artificial intelligence does not operate in isolation. Its effectiveness depends on its integration with sensing and data acquisition systems, computational infrastructure, and physical execution layers such as control systems, robotics, and energy platforms. In this sense, the focus is on enabling adaptive, data-driven, and autonomous industrial systems through the convergence of machine learning, digital twins, and intelligent decision-making frameworks. This perspective represents a key differentiator, as it shifts the emphasis from standalone AI models to fully integrated industrial applications.

The area of *Supply Chain & Logistics Systems* addresses the design, optimization, and transformation of end-to-end supply chains. It integrates artificial intelligence, optimization methods, and empirical data to tackle challenges related to resilience, efficiency, and sustainability. Special attention is given to the increasing importance of Scope 3 emissions, regulatory pressures, and the need for more adaptive and transparent supply chain systems in a highly volatile global environment.

Finally, the *Low-Income Firms Transformation Lab (LIFT Lab)* focuses on the application of artificial intelligence to support micro and small enterprises, particularly in low-data environments. Building on its development at the Massachusetts Institute of Technology (MIT) Center for Transportation & Logistics (CTL) and its ongoing expansion at Tecnológico de Monterrey, this initiative combines technological innovation with large-scale field deployment, enabling measurable impact in productivity, inclusion, and resilience. This area complements the broader industrial focus of AIMS by addressing critical societal challenges linked to supply chains and economic development.

3. Relevance for Industry Engagement

The structure and focus of AIMS respond directly to a clear gap in industry needs. While organizations have access to significant advances in artificial intelligence, materials science, and operations research, these capabilities are often developed and applied in isolation. As a result, companies face challenges in translating technical advances into integrated, deployable solutions that can operate effectively in real-world environments.

AIMS is specifically designed to bridge this gap by enabling the integration of capabilities across domains and by emphasizing deployment as a core component of research. This approach aligns with the increasing demand from industry for solutions that are not only technically robust but also scalable, implementable, and capable of delivering measurable outcomes. In this sense, the Institute's focus is not only academically relevant but also directly connected to high-priority challenges faced by industry.

4. Institutional Role

Within Tecnológico de Monterrey, AIMS will operate as a unifying platform that brings together faculty, capabilities, and infrastructure across schools and campuses to address complex industrial challenges. Its role is not to replace or duplicate existing disciplinary strengths, but rather to integrate and extend them, enabling coordinated efforts that can achieve greater scale and impact.

In short, AIMS will operate as:

- A platform to convene top faculty across schools
- A mechanism to scale industry-sponsored research
- A bridge between research, deployment, and impact

It does not replace disciplinary capabilities, instead, it integrates and amplifies them. By facilitating industry-sponsored research, cross-disciplinary collaboration, and technology deployment, AIMS strengthens the institution's ability to engage with external partners and to position itself as a leader in the transformation of manufacturing and supply chain systems. This institutional role is central to the VP of Research's long-term vision that contributes to the research goals of Grupo Educativo Tecnológico de Monterrey.